ELEC40013 - MATHEMATICS1A - SOLUTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. Given complex $z = x + iy$, which of the following are the real and imaginary parts of $w = \sin(iz)$? (See also Section B, Q1a.)

   (a) $\mathrm{Im}(w) = \sin y \cosh x$, and $\mathrm{Re}(w) = -\cos y \sinh x$,

   (b) $\mathrm{Re}(w) = -\cos y \cosh x$, and $\mathrm{Im}(w) = \sin y \sinh x$,

   (c) $\mathrm{Im}(w) = \sin x \cosh y$, and $\mathrm{Re}(w) = -\cos x \sinh y$,

   (d) $\mathrm{Re}(w) = -\sin y \cosh x$, and $\mathrm{Im}(w) = \cos y \sinh x$,

   (e) none of the above. [3]

   Solution Begin with
   $$\sin(iz) = \sin(ix - y) = \sin(ix)\cos y - \cos(ix)\sin y$$
   Now use relationship between trigonometric and hyperbolic functions to write
   $$\sin(iz) = i\sinh x \cos y - \cosh x \sin y$$
   giving real and imaginary parts as in (d).

2. Which of the following accurately describes the locus in the complex plane satisfying $|z - i| < |z + 2|$?

   (a) the area above the line $y = -2x - \dfrac{3}{2}$, (b) the area below the line $y = 2x - \dfrac{1}{2}$,

   (c) the area below the line $x = -2y + \dfrac{1}{2}$, (d) the area above the line $x = 2y + \dfrac{3}{2}$,

   (e) none of the above. [3]

   Solution Consider equal distance:

   $$|z - i| = |z + 2| \Rightarrow \sqrt{x^2 + (y-1)^2} = \sqrt{(x+2)^2 + y^2} \Rightarrow -2y + 1 = 4x + 4 \Rightarrow y = -2x - \frac{3}{2},$$

   so the only possible solution is (a). As the distance to $z = i$ is smaller than the distance to $z = -2$, we require the area above the line, confirming (a) as the solution.

3. Which of the following statements about $\lim_{x\to\infty} \dfrac{\sqrt{2x+1}-\sqrt{2x-1}}{x^{-1/2}}$ is true?

(a) the limit does not exist, (b) the limit is equal to 0,

(c) the limit is equal to $\dfrac{1}{\sqrt{2}}$, (d) the limit cannot be determined,

(e) none of the above. [3]

Solution: First write $\sqrt{2x\pm 1} = \sqrt{2x\left(1\pm\dfrac{1}{2x}\right)}$ and rewrite

$$\frac{\sqrt{2x+1}-\sqrt{2x-1}}{x^{-1/2}} = \sqrt{2}\,x^{(1/2+1/2)}\left[\left(1+\frac{1}{2x}\right)^{1/2}-\left(1-\frac{1}{2x}\right)^{1/2}\right]$$

$$= \sqrt{2}\,x\left[\left(1+\frac{1}{2}\left(\frac{1}{2x}\right)+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left(\frac{1}{2x}\right)^2+\ldots\right)-\left(1+\frac{1}{2}\left(-\frac{1}{2x}\right)+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left(-\frac{1}{2x}\right)^2+\ldots\right)\right]$$

$$= \sqrt{2}\,x\left(\frac{1}{2x}+\frac{K_1}{x^3}+\frac{K_2}{x^5}+\ldots\right) = \frac{\sqrt{2}}{2}\left(1+\frac{K_3}{x^2}+\frac{K_4}{x^4}+\ldots\right)$$ for constants K_i. Hence, as $x\to\infty$, the limit exists and is equal to $\sqrt{2}/2$, hence (c).

4. The limit given as $\lim_{x\to -1} \dfrac{1-\cos(x+1)}{(x+1)^2}$

(a) exists, but cannot be determined, (b) is equal to 0,

(c) does not exist, (d) is equal to $\dfrac{1}{2}$,

(e) none of the above. [3]

Solution Check that it's "$0/0$" and use l'Hopital:

$$\lim_{x\to -1} \frac{1-\cos(x+1)}{(x+1)^2} = \lim_{x\to -1} \frac{\sin(x+1)}{2(x+1)}$$

which is still "$0/0$", so once more:

$$= \lim_{x\to -1} \frac{\cos(x+1)}{2} = \frac{1}{2}, \text{ hence (d).}$$

5. Let two functions be $f(x) = \ln(x + \sqrt{x^2+1})$ and $g(x) = \dfrac{e^{-x}\ln x}{x}$. Then

(a) $f'(x) = \dfrac{1}{\sqrt{x^2+1}}$, and $\dfrac{dg}{dx} = \dfrac{e^{-x}[(\ln x - 1)x - 1)}{x^2}$,

(b) $\dfrac{dg}{dx} = \dfrac{1}{\sqrt{x^2+1}}$, and $f'(x) = \dfrac{e^{-x}[(1-(x+1)\ln x]}{x^2}$,

(c) $\dfrac{df}{dx} = \dfrac{1}{\sqrt{x^2-1}}$, and $g'(x) = \dfrac{e^{-x}[\ln x(x-1)+1]}{x^2}$,

(d) $f'(x) = \dfrac{x}{\sqrt{x^2+1}}$, and $g'(x) = \dfrac{e^{-x}(x - \ln x + 1)}{x^2}$,

(e) none of the above. [2]

Solution

$$f'(x) = \frac{1}{x+\sqrt{x^2+1}}\left(1 + \frac{2x}{2\sqrt{x^2+1}}\right) = \frac{1}{\sqrt{x^2+1}},$$

and

$$g'(x) = \frac{\left(\frac{e^{-x}}{x} - e^{-x}\ln x\right)x - e^{-x}\ln x}{x^2} = \frac{e^{-x}[(1-(x+1)\ln x]}{x^2},$$

all answers incorrect for various errors, so (e).

6. Which of the following statements about the function $g(x) = \dfrac{e^{-x}\ln x}{x}$ is true? (See also Section B, Q1b.)

(a) there is a local maximum at x_1, between 0 and 1,

(b) as $x \to 0$, $g(x) \to \infty$, so the $y-$axis is an asymptote,

(c) as $x \to \infty$, $g(x) \to 0^-$, so the $x-$axis is an asymptote,

(d) there is a point of inflection at $x_2 > 1$

(e) none of the above. [2]

Solution: We have the derivative from Q5, but can reason without it: the function is only defined for $x > 0$. As $x \to 0^+$, we have $1/x \to +\infty$ and $\ln x \to -\infty$; as e^{-x} is always positive, we get $g(x) \to -\infty$. At $x = 1$, we cross the axis, and nowhere else, so further to the right, as $x \to \infty$, the exponential term takes $g(x)$ to zero, from above. There must be a maximum, somewhere beyond $x = 1$, at $x_1 > 1$, so there will be a point of inflection at $x_2 > x_1 > 1$, hence (d).

7. If $x = 1 - \cos v$, and $y = \sin v - v$, where v is a parameter, which of the following is $\dfrac{dy}{dx}$?

(a) $\cot\left(\dfrac{v}{2}\right)$, (b) $-\tan\left(\dfrac{v}{2}\right)$, (c) $\cot v + \csc v$, (d) $\dfrac{\cos v + 1}{\sin v}$,

(e) none of the above. [2]

Solution

$$\frac{dy}{dv} = \cos v - 1 \quad \text{and} \quad \frac{dx}{dv} = \sin v \Rightarrow \frac{dy}{dx} = \frac{\cos v - 1}{\sin v} = \frac{-2\sin^2(v/2)}{2\sin(v/2)\cos(v/2)},$$

hence (b).

8. If $I(x) = \displaystyle\int_0^x te^{-t}\,dt$ is a function of x, which of the following statements is **false**?

(a) $I(x)$ increases most rapidly when $x = 1$, (b) $I(x)$ has a point of inflection at $x = 2$,

(c) $I(x)$ has a stationary point at $x = 0$, (d) $I(x)$ is increasing for all $x > 0$,

(e) none of the above . [3]

Solution If $I(x)$ is the integral, then the fundamental theorem of calculus gives $\dfrac{dI}{dx} = xe^{-x} = 0$ when $x = 0$, so a stationary point: (c) is true. This is also the rate of increase of I, which is positive for all $x > 0$, so (d) is true. Differentiate: $\dfrac{d^2I}{dx^2} = (1-x)e^{-x} = 0$ when $x = 1$, so the rate of increase is maximal at $x = 1$: (a) is true. This is also the only point of inflection for I, so (b) is false.

9. The integral $I = \displaystyle\int \frac{1}{\cos x - 1}\,dx$ is equal to

(a) $\cot(x/2)$+C, (b) $\cot x - \csc x$+C, (c) $\dfrac{x}{\cos x + 1}$+C, (d) $\dfrac{1}{\tan(x/2) - 1}$+C,

(e) none of the above. [3]

Solution Multiply integrand appropriately

$$I = \int \frac{1}{\cos x - 1}\frac{1+\cos x}{1+\cos x}\,dx = -\int \frac{1}{\sin^2 x}\,dx - \int \frac{\cos x}{\sin^2 x}\,dx = \cot x + \csc x + C$$

or, better, use substitution $t = \tan(x/2), dx = \dfrac{2\,dt}{1+t^2}, \cos x = \dfrac{1-t^2}{1+t^2}$, so

$$I = \int \frac{1}{\left(\frac{1-t^2}{1+t^2} - 1\right)}\frac{2\,dt}{(1+t^2)} = \int \frac{2}{-2t^2}\,dt = \frac{1}{t} + C = \frac{1}{\tan(x/2)} + C = \cot(x/2) + C$$

so (a)

10. Two series are given as

$$S_1 = \sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{and} \quad S_2 = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}} \, .$$

If R means radius of convergence, which of the following statements is **false**?

(a) S_1 has $R = 1$, and S_2 converges conditionally,

(b) S_2 is an alternating series, and S_1 converges for $x = -1$,

(c) S_1 diverges for $x = 1$, and S_2 has $R = 1$,

(d) S_2 converges, but not by the integral test, and S_1 diverges for $|x| > 1$,

(e) none of the above. [3]

Solution S_1 has radius of convergence $R = 1$, using d'Alembert's ratio test. S_2 coverges, using the alternating series test, but taking the absolute value of the summand, we have divergence, use the integral test, so this converges conditionally. Letting $x = -1$ in S_1 gives the alternating harmonic series, which converges, use the alternating series test; letting $x = 1$, we get the harmonic series which diverges, use the integral test. The integral test is no use to show convergence of an alternating series like S_2, as it requires the terms of the sequence to be of the same sign. The ratio test shows divergence for S_1 when $x > 1$ or $x < -1$. Hence, all are true, except for S_2 which is not a power series, has no radius of convergence. Hence (c).

11. Which of the following are the first three terms of the power series of $f(x) = \sqrt{1-x}$?

(a) $1 - \frac{1}{2}x - \frac{1}{4}x^2$, (b) $1 - \frac{1}{2}x - \frac{1}{8}x^2$, (c) $1 + \frac{1}{2}x + \frac{1}{4}x^2$, (d) $1 - \frac{1}{2}x + \frac{1}{8}x^2$,

(e) none of the above. [2]

Solution Using the binomial expansion $(1-x)^n = 1 + n(-x) + \frac{n(n-1)}{2}(-x)^2 + \cdots$, is easiest, with $n = 1/2$ giving

$$(1-x)^{1/2} = 1 - \frac{1}{2}x + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2}x^2 + \cdots = 1 - \frac{1}{2}x - \frac{1}{8}x^2 + \cdots$$

so (b).

12. In the previous question, we obtained the first three terms of the Maclaurin series of $f(x) = \sqrt{1-x}$. Which of the following is the correct remainder term R_2?

(a) $R_2 = -\frac{x^3}{16(1-c)^{5/2}}$, with $0 < c < x$, or $x < c < 0$,

(b) $R_2 = -\frac{1}{4(1+c)^{3/2}}x^2$, with $c < x < 0$ or $x > c > 0$,

(c) $R_2 = -\frac{3x^3}{8(1+c)^{5/2}}$, with $0 < |c| < x$,

(d) $R_2 = \dfrac{1}{8(1-c)^{3/2}}x^2$, with $0 < x < c$ or $x > c > 0$,

(e) none of the above. [3]

Solution Differentiating, we get

$$f'(x) = -\frac{1}{2}(1-x)^{-1/2} \Rightarrow f''(x) = -\frac{1}{4}(1-x)^{-3/2} \Rightarrow f'''(x) = -\frac{3}{8}(1-x)^{-5/2}\,,$$

and from the formula sheet,

$$R_2 = \frac{f'''(c)}{3!}x^3 \;=\; -\frac{x^3}{16(1-c)^{5/2}}$$

where $0 < |c| < |x|$, hence (a).

13. To solve the first order ordinary differential equation (ODE)

$$x^2\frac{dy}{dx} - xy + y^2 = 0$$

we begin by rewriting in terms of $v = \dfrac{y}{x}$, to obtain the following separable first-order ODE:

(a) $\dfrac{dv}{dx} = -v^2 + v$, (b) $x\dfrac{dv}{dx} = -v^2$, (c) $-\dfrac{1}{v-v^2}\dfrac{dv}{dx} = \dfrac{1}{x}$, (d) $\dfrac{1}{x}\dfrac{dv}{dx} = -\dfrac{1}{v^2}$,

(e) none of the above . [3]

Solution Begin by rewriting as

$$\frac{dy}{dx} - \frac{y}{x} + \left(\frac{y}{x}\right)^2 \;=\; 0\,,$$

and with

$$y = xv \Rightarrow \frac{dy}{dx} \;=\; x\frac{dv}{dx} + v$$

we get

$$x\frac{dv}{dx} - v + v + v^2 = 0\,,$$

hence (b).

14. The solution of the homogenous first-order ODE

$$x^2\frac{dy}{dx} - xy + y^2 = 0$$

is obtained from the separable ODE found in the previous question, with $C \in \mathbb{R}$, as follows:

(a) $y \;=\; -\dfrac{x}{C+\ln x}$, (b) $y \;=\; \dfrac{\ln(Cx)}{x}$, (c) $y \;=\; \dfrac{x}{\ln(Cx)}$, (d) $y \;=\; -x(C+\ln x)$,

(e) none of the above. [3]

Solution Begin with the separable ODE

$$x\frac{dv}{dx} = -v^2 \Rightarrow -\frac{1}{v^2}\frac{dv}{dx} = \frac{1}{x} \Rightarrow \frac{1}{v} = \frac{x}{y} = \ln x + K$$

where K is a constant. Hence $\dfrac{y}{x} = \dfrac{1}{\ln(Cx)}$, where $\ln C = K$, giving answer (c).

15. A function of two variables is given as $f(x,y) = e^{xy}\sin x$. Which of the following are correct first partial derivatives?

(a) $f_x = xe^{xy}\cos x$, and $f_y = e^{xy}(x\sin x + \cos x)$,

(b) $f_y = xe^{xy}\sin x$, and $f_x = e^{xy}(\sin x - y\cos x)$,

(c) $f_y = ye^{xy}\cos x$, and $f_x = e^{xy}(y\sin x + \cos x)$,

(d) $f_x = ye^{xy}\sin x$, and $f_y = e^{xy}(x\cos x - \sin x)$,

(e) none of the above. [2]

Solution Correctly, $f_x = e^{xy}(y\sin x + \cos x)$, as in (c), and $f_y = xe^{xy}\sin x$, as in (b), but not together, so (e)

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. (a) Obtain all complex solutions of the equation $\sin(iz) = -2$. Hence, or otherwise, obtain all complex solutions of $\sinh^2(z) = -4$. [Hint: see also Section A, Q1.] [7]

Solution Following the hint, we had earlier found

$$\sin(iz) = i\sinh x\cos y - \cosh x\sin y = -2$$

Hence, comparing real and imaginary parts, we get

$$\sinh x\cos y = 0 \quad (i) \quad \text{and} \quad \cosh x\sin y = 2 \quad (ii)$$

Equation (i) gives $\sinh x = 0 \Rightarrow x = 0$, or $\cos y = 0 \Rightarrow y = (2n+1)\frac{\pi}{2}$, for any integer n. These need to satisfy equation (ii), so begin with $x = 0 \Rightarrow \sin y = 2$ which has no solutions. From $y = (2n+1)\frac{\pi}{2} \Rightarrow \pm\cosh x = 2$, we have $x = \pm\cosh^{-1}(2)$, when $\sin y = 1$, leaving $y = \frac{\pi}{2} + 2n\pi$, for any integer n. Hence

$$z = \pm\cosh^{-1}(2) + i\left(\frac{\pi}{2} + 2n\pi\right) .$$

A lot of errors, most commonly forgetting $\cosh x > 0$*, so allowing* $\sin y = \pm 1$*, not restricting the values for y. Many got the restriction right but then took only* $x = \cosh^{-1}(2)$*, missing the negative value. Many went straight from* $\sin y = \pm 1$ *to* $x = \cosh x = \pm 2$ *and concluded* $x = \cosh^{-1}(\pm 2)$*. Many tried using exponentials at one step or another, which invariably led to errors.*

For the second part, need to spot that using $\sinh z = -i\sin(iz)$ may be useful, rather than starting from scratch. Begin with

$$\sinh^2(z) = -4 \Rightarrow \sinh z = \pm 2i \Rightarrow \sin(iz) = \pm 2,$$

so we already have half the answers. To deal with $\sin(iz) = 2$ we go back to equation (ii) and again, $x = 0$ gives no solutions, while $y = (2n+1)\frac{\pi}{2}$ gives the same solutions for $x = \pm\cosh^{-1}(2)$, but this time we need $\sin y = -1$, so here $y = -\frac{\pi}{2} + 2n\pi$. Putting the two cases together we now have

$$z = \pm\cosh^{-1}(2) + i\left(\frac{\pi}{2} + n\pi\right) .$$

Many tried this completely independently of the first part, sometimes converting $\sinh z$ *to exponentials, and in some cases squaring this, leading to lots of errors. Of those that saw the connection to the first part, few realized the implication. Many took the root incorretly, getting* $\sinh z = 2i \Rightarrow \sin(iz) = 2$*, giving the same answer.*

(b) Given $g(x) = \dfrac{e^{-x}\ln x}{x}$, obtain all stationary points and their nature, and show that the interval $[1,2]$ contains a stationary point. Obtain all asymptotes, intercepts, points of inflection, and hence sketch $g(x)$, explaining your reasoning. [Hint: See also Section A, Q5&6.] [7]

Solution: Following the hint, we already noted in section A:
(i) the function is only defined for $x > 0$. As $x \to 0^+$, we have $1/x \to +\infty$ and $\ln x \to -\infty$; as e^{-x} is always positive, we get $g(x) \to -\infty$. At $x = 1$, we cross the axis, and nowhere else, so further to the right, as $x \to \infty$, the exponential term takes $g(x)$ to zero, from above. Hence there must be a maximum, somewhere to the right of $x = 1$, and a point of inflection to the right of that.

(ii) The derivative, from A5, is

$$g'(x) = \frac{e^{-x}[(1-(x+1)\ln x]}{x^2} = 0 \Rightarrow 1 = (x+1)\ln x\ .$$

Obviously, when $x = 1$, we have $(1+1)\ln 1 = 0 < 1$. We need to show that for $x = 2$, we get $(2+1)\ln 2 > 1$, in which case, the maximum lies in the desired interval. Given $e = 2.718... \Rightarrow e/2 < 2$, we estimate that

$$\ln(e/2) < \ln 2 \Rightarrow \ln e - \ln 2 < \ln 2 \Rightarrow \ln 2 > 1/2\ .$$

Hence $(2+1)\ln 2 > 3/2 > 1$, as required, the maximum at x_1 is in the interval $[1,2]$. Finally, the curvature changes to the right of the maximum at x_1, as there is no other $x-$intercept, and $g(x) \to 0$ as $x \to \infty$. So we can infer the existence, but not location of a point of inflection at $x_2 > x_1$, as seen in A6. Sketch at end.

Unexpectedly, the most difficult question on the exam, from the answers given. Almost everyone got the correct derivative. Of these, not everyone saw how to show that the interval $[1,2]$ *contains a stationary point. The most comon correct version seen was taking the derivative as the positive* e^{-x}/x^2 *multiplying*

$$h(x) = 1-(x+1)\ln x\ ,$$

so $h(1) = 1 > 0$ *and* $h(2) = 1 - 3\ln 2 < 0$*, giving a change of sign in-between.*
An interesting alternative came as

$$(x+1)\ln x = 1 \Rightarrow \ln\left(x^{x+1}\right) = 1 \Rightarrow h(x) = x^{x+1} = e\ ,$$

with $h(x)$ *an increasing function, and as* $h(1) = 1 < e$ *and* $h(2) = 8 > e$*, we can rightly conclude that for some* $1 < x_0 < 2$*,* $h(x_0) = e$*, giving a stationary point.*
Others drew the curves $1/(x+1)$ *and* $\ln x$ *and correctly identified that to the right of* $x = 1$*, these intersect, though not always making the argument that this is to the left of* $x = 2$*.*
Most people tried the second derivative, but it was no use, as we don't know the value of the SP. Many had the vertical asymptote to $+\infty$*, then crossing the* $x-$*axis at* $x = 1$ *and sketching a minimum, and approaching the horizontal asymptote from below, even though many of these had correctly decided earlier that the SP was a max, and the asymptote is approached from above. Some then drew a max to the right of the min. Few thought about the POI. Many drew some curve for* $x < 0$*.*

(c) Given $g(x) = \dfrac{e^{-x}\ln x}{x}$ as in Q1b, let $f(x) = g(x)[H(x) - H(x-2)]$, where $H(x)$ is the Heaviside function. Sketch $f(x)$, and hence sketch $f_e(x)$ and $f_o(x)$, the even and odd parts of $f(x)$, taking care to label the axes. [6]

Solution We observe that the effect of the Heaviside terms is to take a window of $g(x)$, on the interval $[0,2]$. We can sketch $f(x)$, then reflect into $[-2,0]$ for $f(-x)$ and add/subtract to obtain even and odd parts. Need to label the $y-$axis so that for $f(x)$, the value at the stationary point is clearly twice that for the even/odd parts, see sketch at end.

Many got $f(x)$ right, at least matching the $g(x)$ from (b), but then couldn't draw even/odd parts. Many of these wrote down formulae for even/odd part, even with the function, so very complicated, but these led nowhere. Of those that got even and odd part matching the $f(x)$ correctly, almost everyone forgot to include the fact that the max/min values are half the max of $f(x)$.

2. (a) Obtain the integrals: $\int \sinh^{-1} x\, dx$ and $\int \frac{1}{(1+x^2)^2}\, dx$. [7]

Solution (i) From the formula sheet we obtain $\frac{d}{dx}\left(\sinh^{-1} x\right) = \frac{1}{\sqrt{1+x^2}}$. So for $I = \int \sinh^{-1} x\, dx$ we integrate by parts, with $u = \sinh^{-1} x \Rightarrow du = \frac{1}{\sqrt{1+x^2}} dx$ and $dv = dx \Rightarrow v = x$ so that

$$I \;=\; x\sinh^{-1} x - \int \frac{x}{\sqrt{1+x^2}}\, dx \;=\; x\sinh^{-1} x - \sqrt{1+x^2} + C\,.$$

Most people got this. A few got the last integration to give $\ln(\sqrt{1+x^2})$*, or similar.*

(ii) Rewrite as

$$I = \int \frac{1+x^2-x^2}{(1+x^2)^2}\, dx \;=\; \int \frac{1}{1+x^2} - \frac{x^2}{(1+x^2)^2}\, dx \;=\; \tan^{-1} x - \int \frac{x^2}{(1+x^2)^2}\, dx$$

For the second integral, we use integration by parts, letting

$$u = x \Rightarrow du = dx \quad \text{and} \quad dv = -\frac{x}{(1+x^2)^2} dx \Rightarrow v = -\frac{1}{2(1+x^2)}$$

so

$$\int \frac{x^2}{(1+x^2)^2}\, dx \;=\; -\frac{x}{2(1+x^2)} + \frac{1}{2}\int \frac{1}{1+x^2}\, dx \;=\; -\frac{x}{2(1+x^2)} + \frac{1}{2}\tan^{-1} x$$

and finally

$$I = \tan^{-1} x - \left(-\frac{x}{2(1+x^2)} + \frac{1}{2}\tan^{-1} x\right) = \frac{1}{2}\left(\frac{x}{1+x^2} + \tan^{-1} x\right) + C$$

Few right anwers. Interestingly, an alternative approach takes the substitution $x = \cos u$ *giving the alternative* $\frac{1}{4}\sin(2\tan^{-1} x) + \frac{1}{2}\tan^{-1} x$*. A lot tried integration by parts, with* $u = \frac{1}{1+x^2}$ *,*$dv = \frac{1}{1+x^2}$ *, or similar, leading nowhere.*

(b) Find a recurrence relation for the integral $I_n = \int_0^{\pi/2} \sin^n(2x)\, dx$, and use this to obtain I_4 and I_5. [6]

Solution Integration by parts: let

$$u = \sin^{n-1}(2x) \Rightarrow du = (n-1)\sin^{n-2}(2x)2\cos(2x)\, dx \quad \text{and} \quad dv = \sin(2x)dx \Rightarrow v = -\frac{1}{2}\cos(2x)\,,$$

so

$$I_n = -\frac{1}{2}\cos(2x)\sin^{n-1}(2x)\Big]_0^{\pi/2} + \int_0^{\pi/2}(n-1)\sin^{n-2}(2x)2\cos(2x)\frac{1}{2}\cos(2x)\, dx$$

$$= (n-1)\int_0^{\pi/2}\sin^{n-2}(2x)\cos^2(2x)\, dx = (n-1)\int_0^{\pi/2}\sin^{n-2}(2x)(1-\sin^2(2x)\, dx = (n-1)(I_{n-2}-I_n)\,,$$

giving the recurrence relation $I_n = \dfrac{n-1}{n} I_{n-2}$. The first two are easy:

$$I_0 = \int_0^{\pi/2} 1\, dx = \frac{\pi}{2} \Rightarrow I_2 = \frac{1}{2}I_0 = \frac{\pi}{4} \Rightarrow I_4 = \frac{3}{4}I_2 = \frac{3\pi}{16}\,,$$

and

$$I_1 = \int_0^{\pi/2}\sin(2x)\, dx = 1 \Rightarrow I_3 = \frac{2}{3}I_1 = \frac{2}{3} \Rightarrow I_5 = \frac{4}{5}I_3 = \frac{8}{15}\,.$$

Mostly done well. A lot of errors such as $du = 2(n-1)\sin^{n-2}(2x)$, and then running into difficulty in the next step, having only $\cos(2x)$ in the integrand. Or letting $dv = dx \Rightarrow v = x$, and having x in the integrand in the next step. A few people wrote

$$I_n = \int_0^{\pi/2}\sin^{n-2}(2x)[1-\cos^2(2x)]\, dx = I_{n-2} - \int_0^{\pi/2}\sin^{n-2}(2x)\cos^2(2x)\, dx$$

which can then be integrated by parts, with $u = \cos 2x$ and $dv = sin^{n-2}(2x)\cos 2x\, dx$, usually correctly.

(c) Let a linear, homogenous second-order ODE with constant coefficients

$$\frac{d^2x}{dt^2} + A\frac{dx}{dt} + Bx = 0 \qquad (*)$$

have A, B real. Let $y_1(t)$ and $y_2(t)$ be real functions of t, and $i = \sqrt{-1}$. Prove that if $z = y_1 + iy_2$ is a solution of $(*)$, then y_1 and y_2 are also solutions. [7]

Solution: Given that z is a solution, we differentiate twice and substitute into $(*)$:

$$z = y_1 + iy_2 \Rightarrow z' = y_1' + iy_2' \Rightarrow z'' = y_1'' + iy_2''$$

so

$$y_1'' + iy_2'' + A(y_1' + iy_2') + B(y_1 + iy_2) = 0 \Rightarrow y_1'' + Ay_1' + By_1 + i(y_2'' + Ay_2' + By_2) = 0\,.$$

Given that $A, B \in \mathbb{R}$ and y_1, y_2 are real functions, the last expression is the sum of two real quantities, one multiplied by i, so we can equate real and imaginary parts, to get

$$y_1'' + Ay_1' + By_1 = 0, \quad \text{and} \quad y_2'' + Ay_2' + By_2 = 0\,.$$

Hence y_1 and y_2 are also solutions of $(*)$, as required.

Not many correct proofs. A lot errors involving complicated manipulations of an auxiliary equation in A,B, and what this means for complex conjugate solutions with exponentials and $\cos + i\sin$. *One person managed to show that the resulting real and imaginary parts are indeed also solutions, but most got lost at this point. Of those that had the proof correct, only one person mentioned that A,B needed to be real.*

3. (a) Use the Leibnitz-Maclaurin method to find the first four non-zero terms of the series solution of the second-order ODE

$$(1+t^2)\frac{d^2x}{dt^2}+2t\frac{dx}{dt}=0 \qquad (*)$$

which satisfies the initial conditions $x(0)=x'(0)=1$. Begin by differentiating the ODE n times, and obtain a recurrence relation involving $x^{(n+2)}(0)$ and $x^{(n)}(0)$. By $x^{(k)}(0)$, we mean x differentiated k times, and then evaluated at $t=0$. [8]

Solution: First, use the Leibnitz formula to differentiate the two products n times:

$$\frac{d^n}{dt^n}\left((1+t^2)\frac{d^2x}{dt^2}\right) \;=\; (1+t^2)x^{(n+2)}(t)+n(2t)x^{(n+1)}(t)+2\frac{n(n-1)}{2}x^{(n)}(t)\;.$$

Likewise

$$\frac{d^n}{dt^n}\left(t\frac{dx}{dt}\right) \;=\; tx^{(n+1)}(t)+nx^{(n)}(t)\;,$$

and putting them together, we have

$$(1+t^2)x^{(n+2)}(t)+n(2t)x^{(n+1)}(t)+n(n-1)x^{(n)}(t)+2tx^{(n+1)}(t)+2nx^{(n)}(t)=0\,.$$

Now set $t=0$ and collect like terms:

$$x^{(n+2)}(0)+n(n-1)x^{(n)}(0)+2nx^{(n)}(0)=0\Rightarrow x^{(n+2)}(0)=-n(n+1)x^{(n)}(0)$$

is our recurrence relation. We can now take the initial conditions and use the recurrence relation:

$x(0)=x^{(0)}(0)=1$

$x'(0)=x^{(1)}(0)=1$

$n=0:\ x^{(2)}(0)=-0(0+1)x^{(0)}(0)=0$

The recurrence relation is in steps of two, so from here on, all the even derivatives will be zero.

$n=1:\ x^{(3)}(0)=-1(1+1)x^{(1)}(0)=-2$

$n=3:\ x^{(5)}(0)=-3(3+1)x^{(3)}(0)=24$, which is enough terms. The Maclaurin series is therefore

$$x(t)=x(0)+x'(0)t+\frac{x^{(2)}(0)}{2!}t^2+\frac{x^{(3)}(0)}{3!}t^3+\cdots=1+t-\frac{2}{3!}t^3+\frac{24}{5!}t^5+\cdots=1+t-\frac{1}{3}t^3+\frac{1}{5}t^5+\cdots\,.$$

Mostly done well. A surprising number or people obtained the correct recurrence relation and then stopped. Most common error, otherwise, apart from arithmetic slipups, was not including n in the binomial coefficients.

(b) Consider again the ODE from part (a):

$$(1+t^2)\frac{d^2x}{dt^2}+2t\frac{dx}{dt}=0 \qquad (*)$$

Now let $z=x'(t)$ and rewrite $(*)$, as a first-order linear ODE in z. Use an integrating factor to solve this, and hence find the exact solution of $(*)$. [6]

Solution: Letting $z=x'(t)\Rightarrow z'(t)=x''(t)$ with the initial condition $x'(0)=z(0)=1$ we have

$$(1+t^2)\frac{dz}{dt}+2tz=0\Rightarrow \frac{dz}{dt}+\frac{2t}{1+t^2}z=0\ ,$$

the first order linear ODE in standard form, and the integrating factor is

$$\mu(t)\ =\ e^{\int\frac{2t}{1+t^2}\,dt}\ =\ e^{\ln(1+t^2)}=1+t^2\ .$$

Multiplying back, we have

$$(1+t^2)\frac{dz}{dt}+2tz\ =\ \frac{d}{dt}\left[\mu(t)z\right]\ =\ \frac{d}{dt}\left[(1+t^2)z\right]=0$$

and integrating we have the solution $(1+t^2)z=C$. The initial condition $z(0)=1$ gives $C=1$ and we have

$$z\ =\ \frac{dx}{dt}\ =\ \frac{1}{1+t^2}\Rightarrow x(t)\ =\ \int\frac{1}{1+t^2}\,dt\ =\ \tan^{-1}t+C$$

The last initial condition $x(0)=1$ gives $C=1$ and the solution is

$$x(t)\ =\ 1+\tan^{-1}t\ .$$

Very well done, though a surprising number of people got to $z=1/(x^2+1)$ and stopped. Many people didn't use the initial conditions to find the values of the arbitrary constants. A common error was integrating

$$(1+x^2)z=\int 0\,dx=x,\ \textit{or}\ x+c$$

.

(c) Given $x(u,v) = u^2 + v^2$ and $y(u,v) = 2uv$, and $f(x,y)$ is an arbitrary function of x and y, show that

$$u\frac{\partial f}{\partial u} - v\frac{\partial f}{\partial v} \;=\; 2(u^2 - v^2)\frac{\partial f}{\partial x}$$

and find a similar expression involving $\dfrac{\partial f}{\partial y}$. [6]

Solution: Using the chain rule,

$$uf_u \;=\; u[f_x x_u + f_y y_u] \;=\; u[f_x(2u) + f_y(2v)], \quad \text{and} \quad vf_v \;=\; v[f_x x_v + f_y y_v] \;=\; v[f_x(2v) + f_y(2u)].$$

Subtracting we get

$$uf_u - vf_v \;=\; 2u^2 f_x - 2v^2 f_x \;=\; 2(u^2 - v^2)f_x,$$

as required. The cancellation suggests obtaining

$$vf_u \;=\; v(f_x x_u + f_y y_u) \;=\; v(f_x(2u) + f_y 2v) \quad \text{and} \quad uf_v \;=\; u(f_x x_v + f_y y_v) \;=\; u(f_x(2v) + f_y 2u)\,,$$

and subtracting again, to get

$$vf_u - uf_v \;=\; 2v^2 f_y - 2u^2 f_y \;=\; 2(v^2 - u^2)f_y.$$

A lot of errors, mostly with people trying to use the much more difficult chain rule:

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x}\,, \qquad (**)$$

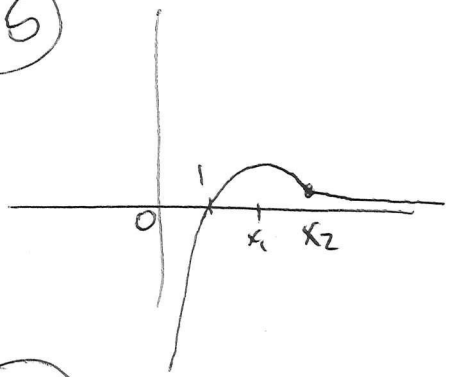
and similar for $\dfrac{\partial f}{\partial y}$. *A lot of people went for* $\dfrac{\partial x}{\partial u} = 2u \Rightarrow \dfrac{\partial u}{\partial x} = \dfrac{1}{2u}$, *etc. A surprising number used*

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{\partial f}{\partial v}\frac{\partial v}{\partial y}\,,$$

or similar, even though the correct chain rule is on the formula sheet. A few people found $u = \dfrac{1}{2}(\sqrt{x+y} + \sqrt{x-y})$ *and* $v = \dfrac{1}{2}(\sqrt{x+y} - \sqrt{x-y})$. *Some of them differentiated correctly to find* u_x, u_y, v_x, v_y. *Some of these then used* $(**)$ *correctly to find* f_x, f_y, *but it was usually a struggle, and the few that succeeded with this were the people who ended up with very high overall marks.*

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(15)



Max at $1 < x_1 < 2$
Pol at $x_2 > x_1$

(16)

